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United States Patent	12408328
Kind Code	B2
Date of Patent	September 02, 2025
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Semiconductor device and method of fabricating the same

Abstract

The present disclosure provides a semiconductor device and a method of fabricating the same, the semiconductor device includes a substrate, an insulating layer, a plurality of bit lines, and a bit line contact. The insulating layer is disposed on the substrate, the bit lines are disposed on the insulating layer, and the bit line contact is disposed between the bit lines and the substrate, to electrically connect the bit lines, wherein the bit line contact comprises a first conductive layer and a first oxidized interface layer, and a bottommost surface of the first oxidized interface layer is lower than a top surface of the insulating layer. Through these arrangements, the semiconductor device includes the bit line contact having a composite semiconductor layer, which is allowable to improve the structural reliability of the bit lines and the bit line contacts, thereby achieve better performance.

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Appl. No.: **17/888525**

Filed: **August 16, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20240040773 A1	Feb. 01, 2024

Foreign Application Priority Data

CN	202210893653.1	Jul. 27, 2022
CN	202221958948.4	Jul. 27, 2022

Publication Classification

Int. Cl.: **H10B12/00** (20230101)

U.S. Cl.:

CPC **H10B12/485** (20230201); **H10B12/315** (20230201); **H10B12/482** (20230201);

Field of Classification Search

CPC: H10B (12/485); H10B (12/315); H10B (12/482)

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present disclosure relates generally to a semiconductor device and a method of fabricating the same, and more particularly to a semiconductor memory device and a method of fabricating the same.

2. Description of the Prior Art

(2) With the trend of miniaturization of various electronic products, the design of semiconductor memory devices must meet the requirements of high integration and high density. For a dynamic random access memory (DRAM) having recessed gate structures, because the carrier channel of which is relatively long in the same semiconductor substrate compared with that of the DRAM without recessed gate structures, the leakage current from the capacitor structure in the DRAM can be reduced. Therefore, the DRAM having recessed gate structures has gradually replaced DRAM having planar gate structures under the current mainstream development trend.

(3) Generally, the DRAM having recessed gate structure is constructed by a large number of memory cells which are arranged to form an array area, and each of the memory cells can be used to store information. Each memory cell may include a transistor element and a capacitor element connected in series, which is configured to receive voltage information from word lines (WL) and bit lines (BL). In order to fulfill the requirements of advanced products, the density of memory cells in the array area must be further increased, which increases the difficulty and complexity of related fabricating processes and designs. Therefore, the present technology needs further improvement to effectively improve the efficiency and reliability of related memory devices.

SUMMARY OF THE INVENTION

(4) An object of the present disclosure is to provide a semiconductor device and a fabricating

method thereof, where a bit line contact having a composite conductive layer is formed to improve the structural reliability of the bit lines and the bit line contact. Then, the semiconductor device may therefore achieve better functions and performance.
(5) To achieve the aforementioned objects, the present disclosure provides a semiconductor device including a substrate, an insulating layer, a plurality of bit lines, and a bit line contact. The insulating layer is disposed on the substrate, and the bit lines are disposed on the substrate. The bit line contact is disposed between the bit lines and the substrate, to electrically connect the bit lines, wherein the bit line contact includes a first conductive layer and a first oxidized interface layer, and a bottommost surface of the first oxidized interface layer is lower than a top surface of the insulating layer.
(6) To achieve the aforementioned objects, the present disclosure provides a method of fabricating a semiconductor device including the following steps. Firstly, a substrate is provided, and an insulating layer is formed on the substrate. Next, a plurality of bit lines is formed on the substrate, and a bit line contact is formed between the bit lines and the substrate, to electrically connect the bit lines, wherein the bit line contact includes a first conductive layer and a first oxidized interface layer, and a bottommost surface of the first oxidized interface layer is lower than a top surface of the insulating layer.
(7) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings are directed to provide a better understanding of the embodiments and are included as parts of the specification of the present disclosure. These drawings and descriptions are used to illustrate the principles of the embodiments. It should be noted that all drawings are schematic, and the relative dimensions and scales have been adjusted for the convenience of drawing. Identical or similar features in different embodiments are marked with identical symbols.
(2) FIG. **1** to FIG. **9** illustrate schematic diagrams of a fabricating method of a semiconductor device according to first embodiment of the present disclosure, wherein:
(3) FIG. **1** is a schematic top view of a semiconductor device after forming a mask layer;
(4) FIG. **2** is a schematic cross-sectional view taken along a cross line A-A′ in FIG. **1**;
(5) FIG. **3** is a schematic top view of a semiconductor device after forming a bit line contact opening;
(6) FIG. **4** is a schematic cross-sectional view taken along a cross line A-A′ in FIG. **3**;
(7) FIG. **5** is a schematic cross-sectional view of a semiconductor device after forming an oxidized interface layer;
(8) FIG. **6** is a schematic cross-sectional view of a semiconductor device after forming another oxidized interface layer;
(9) FIG. **7** is a schematic cross-sectional view of a semiconductor device after forming bit line stacked layers;
(10) FIG. **8** is a schematic top view of a semiconductor device after forming a bit line; and
(11) FIG. **9** is a schematic cross-sectional view taken along a cross line A-A′ in FIG. **8**.
(12) FIG. **10** to FIG. **11** illustrate schematic diagrams of a fabricating method of a semiconductor device according to second embodiment of the present disclosure, wherein:
(13) FIG. **10** is a schematic cross-sectional view of a semiconductor device after forming an oxidized interface layer; and

(14) FIG. **11** is a schematic cross-sectional view of a semiconductor device after forming a bit line.

DETAILED DESCRIPTION

(15) To provide a better understanding of the presented disclosure, preferred embodiments will be described in detail. The preferred embodiments of the present disclosure are illustrated in the accompanying drawings with numbered elements. In addition, the technical features in different embodiments described in the following may be replaced, recombined, or mixed with one another to constitute another embodiment without departing from the spirit of the present disclosure.

(16) Please refer to FIG. **1** to FIG. **9**, which illustrate schematic diagrams of a semiconductor device **100** according to the first embodiment of the present disclosure, with FIG. **1**, FIG. **3**, and FIG. **8** respectively illustrating a top view of the semiconductor device **100** during various fabricating processes, and with FIG. **2**, FIG. **4** to FIG. **7**, and FIG. **9** respectively illustrating a cross-sectional view of the semiconductor device **100** during various fabricating processes. Firstly, as shown in FIG. **1** and FIG. **2**, a substrate **110**, for example a silicon substrate, a silicon containing substrate (such as SiC or SiGe), or a silicon-on-insulator (SOI) substrate, is provided, and at least one shallow trench isolation (STI) **112** is formed in the substrate **110**, to define a plurality of active areas (AAs) **114**, with all of the active areas **114** being surrounded by the shallow trench isolation **112**, as shown in FIG. **1**.

(17) Precisely speaking, the active areas **114** is parallel extended with each other along a same direction Dl, with each of the active areas **114** having the same length and the same pitch, wherein the direction D**1** for example intersects and is not perpendicular to the y direction or the x direction, as shown in FIG. **1**, but not limited thereto. In one embodiment, the formation of the shallow trench isolation **112** is accomplished by firstly performing an etching process to form a plurality of trenches (not shown in the drawings), followed by filling in an insulating material (such as silicon oxide or silicon oxynitride) in the trenches, but is not limited thereto.

(18) Also, at least two regions are defined on the substrate **110**, for example including a memory cell region (not shown in the drawings) being relative higher integrity and a periphery region (not shown in the drawings) being relative lower integrity, and a plurality of buried word lines **120** is formed in the substrate **110**, within the memory cell region, wherein the buried word lines **120** are parallelly extended with each other in the y-direction, to intersect with each of the active areas **114**. In one embodiment, the formation of the buried word lines **120** includes but not limited to the following steps. Firstly, a plurality of trenches (not shown in the drawings) parallelly and separately extended along the y-direction is formed in the substrate **110**. Next, a dielectric layer (not shown in the drawings) covering entire surfaces of each of the trenches, a gate dielectric layer (not shown in the drawings) covering surfaces of a bottom portion of each of the trenches, and a gate electrode (not shown in the drawings) filled in the bottom portion of each of the trenches, are sequentially formed in each of the trenches. Then, after removing the gate electrode and the gate dielectric layer formed in a top portion of each of the trenches through an etching process, a covering layer (not shown in the drawings) filled in the top portion of each of the trenches is formed. Accordingly, the covering layer may therefore have a top surface leveled with a top surface of the substrate **110**, so that, the gate electrode, the gate dielectric layer and the dielectric layer formed in the substrate **110** may together form the buried word lines **120** of the semiconductor device **100** for accepting or transmitting the voltage signals from memory cells (not shown in the drawings).

(19) As shown in FIG. **1** and FIG. **2**, an insulating layer **130**, and a mask layer **140** are sequentially formed on the substrate **110**, entirely covering the top surface of the substrate **110**, wherein the insulating layer **130** preferably includes a composite structure for example including an oxide layer **132**-*a* nitride layer **134**-*an* oxide layer **136** (ONO) structure, but is not limited thereto. The mask layer **140** for example includes a semiconductor layer **142** and a protection layer **144** formed sequentially on the insulating layer **130**, wherein the semiconductor layer **142** for example includes a semiconductor material like doped silicon, doped phosphorus or silicon phosphide (SiP), and preferably includes doped silicon, but not limited thereto. The protection layer **144** for example

includes a material like silicon oxide or silicon oxynitride (SiON), but not limited thereto.
(20) Then, as shown in FIG. **3** and FIG. **4**, a plurality of contact openings **146** is formed in the mask layer **140**, with each of the contact openings **146** penetrating through the protection layer **144**, the semiconductor layer **142**, and the insulating layer **130** to extend into the substrate **110**, and with a portion of the active areas **114** being exposed from the contact openings **146**. The formation of the contact openings **146** includes but not limited to the following steps. Firstly, a mask structure (not shown in the drawings) is formed on the substrate **110**, with the mask structure for example including a sacrificial layer (not shown in the drawings, for example including an organic dielectric layer), a silicon-containing hard mask (SHB, not shown in the drawings), and a patterned photoresist layer (not shown in the drawings) stacked one over another on the protection layer **144**. The patterned photoresist layer includes at least one pattern for defining the contact openings **146**, and an etching process such as a dry etching process is performed through the patterned photoresist layer, to form the contact openings **146** in the insulating layer **130**, the semiconductor layer **142**, and the protection layer **144**, wherein each of the contact openings **146** is in alignment with each of the active areas **114**. It is noted that, each of the contact openings **146** is formed between two adjacent ones of the buried word lines **120** as shown in FIG. **3**, so that, a portion of the active areas **114** (namely, the substrate **110**) may be exposed from the bottom of the contact openings **146**, as shown in FIG. **4**. After that, the mask structure is completely removed.
(21) As shown in FIG. **5**, a first semiconductor material layer **152** is formed on the substrate **110**, entirely covering the mask layer **140**, and surfaces of each contact opening **146** without filling the contact openings **146**, wherein the first semiconductor material layer **152** for example includes a semiconductor material like doped silicon, doped phosphorus, or silicon phosphide, but not limited thereto. It is noted that, in the present embodiment, after forming the first semiconductor material layer **152**, the lattice structure of the surface of the first semiconductor material layer **152** is destroyed by breaking the vacuum or introducing oxygen, so that the lattice structure of a semiconductor material layer deposited on the top in the subsequent process will not continuously grow along the lattice structure of the first semiconductor material layer **152**, so as to obtain a relatively smaller grain size. Accordingly, a first oxidized interface layer **153** is further formed on the surface of the first semiconductor material layer **152**, and the first oxidized interface layer **153** for example includes silicon oxide, phosphorus oxide or silicon phosphorus oxide, and preferably includes silicon phosphorus oxide, but not limited thereto. In one embodiment, since the first oxidized interface layer **153** is conformally formed on the surface of the first semiconductor material layer **152**, an U-shaped cross-section may be obtained thereby within each contact opening **146**, as shown in FIG. **5**.
(22) Furthermore, it is also known that, a thickness of the first oxidized interface layer **153** is quite thin, for example being about angstroms to 1 angstroms, and which may not affect the electrical connection between the bit line contacts and other elements. In one embodiment, the formation of the first semiconductor material layer **152** is for example carried out through a deposition process such as a chemical vapor deposition (CVD) process, and preferably through a selective epitaxial growing (SEG) process, to precisely monitor a thickness T**1** of the first semiconductor material layer **152**, preferably being about 20-50 nanometers (nm), but not limited thereto. In this way, the first oxidize interface layer **153** formed over the first semiconductor material layer **152** may therefore obtain a relative lower forming position within each contact opening **146**, for example being lower than the top surface of the insulating layer **130**. In one embodiment, a bottommost surface B**1** of the first oxidized interface layer **153** is for example lower than the top surface of the insulating layer (namely, the top surface of the oxide layer **136**), even lower than the bottommost surface of the insulating layer **130** (namely, the bottom surface of the oxide layer **132**), as shown in FIG. **5**.
(23) As shown in FIG. **6**, a deposition process such as a chemical vapor deposition is performed, to sequentially form a second semiconductor material layer **154** and a third semiconductor material

layer **156** on the substrate **110**, wherein the second semiconductor material layer **154** entirely covers the first semiconductor material layer **152**, to partially fill in each contact opening **146**, and the third semiconductor material layer **156** further covers on the second semiconductor material layer **154**, to fill up the rest space of each contact opening **146**. The second semiconductor material layer **154** and the third semiconductor material layer **156** for example include a semiconductor material like doped silicon, doped phosphorus, or silicon phosphide, but not limited thereto. In one preferably embodiment, the forming process of the second semiconductor material layer **154** and the third semiconductor material layer **156** may be different from that of the first semiconductor material layer **152**, for example, the second semiconductor material layer **154** and the third semiconductor material layer **156** may be formed through a CVD process, and the first semiconductor material layer **152** maybe formed through a SEG process, but not limited thereto. Also, the first semiconductor material layer **152**, the second semiconductor material layer **154**, and the third semiconductor material layer **156** include the same material like silicon phosphide, but not limited thereto.

(24) As the semiconductor material of the second semiconductor material layer **154** may also generate a second oxidize interface layer **155** through breaking vacuum or introducing oxygen, the second oxidize interface layer **155** may be formed between the second semiconductor material layer **154** and the third semiconductor material layer **156**, and also, to obtain an U-shaped cross-section within each contact opening **146**, as shown in FIG. **6**. The second oxidized interface layer **155** for example includes silicon oxide, silicon nitride, or silicon phosphorus oxide, and preferably includes silicon phosphorus oxide, but not limited thereto. It is noted that, a thickness of the second oxidized interface layer **155** is also quite thin, for example being about 0.01 angstroms to 1 angstroms, and which will not affect the electrical connection between the bit line contacts and other elements formed in the subsequent processes. Furthermore, it is also noted that, the second semiconductor material layer **154** and the third semiconductor material layer **156** stacked sequentially on the first semiconductor material layer **152** preferably include a relative smaller thickness T**2**, for example being about 10-20 nanometers, but not limited thereto. Accordingly, the second semiconductor material layer **154** and the third semiconductor material layer **156** stacked over the first semiconductor material layer **152** may therefore obtain a relative smaller grain size and a relative finer lattice structure, so as to present a further smooth surface.

(25) As shown in FIG. **7**, an etching process such as a dry etching process is performed, to completely remove the third semiconductor material layer **156**, the second semiconductor material layer **154**, and the first semiconductor material layer **152** covered on the top surface of the mask layer **140**, and to further remove the mask layer **140** underneath, with the protection layer **144** being completely removed and with the semiconductor layer **142** being partially removed. Meanwhile, the third semiconductor material layer **156**, the second semiconductor material layer **154**, and the first semiconductor material layer **152** filled in each contact opening **146** are all partially removed, so that, a semiconductor layer **152*a***, a first oxidized interface layer **153*a***, a first conductive layer **154*a***, a second oxidized interface layer **155*a***, and a second conductive layer **156*a*** are sequentially formed within each contact opening **146**, thereby forming a plurality of bit line contacts (BLC) **160** filled in each contact opening **146** respectively, wherein the first oxidized interface layer **153*a*** and the second oxidized interface layer **155*a*** both include an U-shape cross-section as shown in FIG. **7**. Then, the top surface of each of the bit line contacts **160** may be coplanar with the top surface of the etched semiconductor layer **142**, as shown in FIG. **7**.

(26) Following these, a deposition process is then performed to form a barrier material layer **162**, a metal material layer **164**, and a covering material layer **166** stacked sequentially on the semiconductor layer **142**. Precisely speaking, the barrier material layer **162** entirely covers the semiconductor layer **142** and the bit line contacts **160**, to directly in contact with the semiconductor layer **142** and the bit line contacts **160** disposed underneath, and the metal material layer **164** and the covering material layer **166** are sequentially covering on the barrier material layer **162**. In one

embodiment, the barrier material layer **162** for example includes tantalum (Ta) and/or tantalum nitride (TaN), or titanium (Ti) and/or titanium. nitride (TiN), and the metal material layer **164** for example includes a low-resistant metal like aluminum (Al), titanium, copper (Cu), or tungsten (W), and the covering material layer **166** for example includes a dielectric material like silicon oxide, silicon nitride, or silicon oxynitride, but is not limited thereto.
(27) Then, as shown in FIG. **8** and FIG. **9**, a photolithography process is performed, and the semiconductor layer **142**, the barrier material layer **162**, the conductive material layer **164**, and the capping material layer **166** stacked sequentially are patterned through a mask layer (not shown in the drawings), to form a plurality of bit lines **168** extended along the x-direction in the memory cell region of the substrate **110**, across the active areas **114** and the buried word lines **120**. Precisely speaking, each of the bit lines **168** includes a semiconductor layer **142***a*, a barrier layer **162***a*, a metal layer **164***a*, and a capping layer **166***a* stacked sequentially from bottom to top. It is noted that, since the bit line contacts **160** each is formed under the barrier layer **162***a* of a corresponding bit line **168**, the semiconductor layer **142***a* of the corresponding bit line **168** is not illustrated in the cross-sectional view as shown in FIG. **9**, and the semiconductor layer **142***a* of the corresponding bit line **168** is formed at two sides to each bit line contact **160** in the x-direction as being viewed from a top view as shown in FIG. **8**. It is noted that the bit line contacts **160** are further disposed below a portion of the bit lines **168**, so that, the portion of the bit lines **168** may enable to extend into the active areas **114** of the substrate **110**, to directly contact the active areas **114**. The bit line contacts **160** are also patterned during performing the photolithography process, with the semiconductor layer **152***a* and the first oxidized interface layer **153***a* disposed at two sides of each bit line contact **160** being partially removed. Then, each of the bit line contacts **160** may include a composite conductive layer, which includes the semiconductor layer **152***a*, the first oxidized interface layer **153***a*, the first conductive layer **154***a*, the second oxidized interface layer **155***a*, and the second conductive layer **156***a* stacked sequentially, with the first oxidized interface layer **153***a* having a linear cross-section, and with the second oxidized interface layer **155***a* having an U-shaped cross-section, as shown in FIG. **9**.
(28) Furthermore, as shown in FIG. **8** and FIG. **9**, a deposition process and an etching back process are sequentially performed, to form a plurality of bit line spacers **170** on two opposite sidewalls of each of the bit lines **168**, respectively. It is noted that, a portion of the bit line spacers **170** further extends into active areas **114** of the substrate **110** to cover on sidewalls of each of the bit line contacts **160**, and the portion of the bit line spacers **170** may directly contact two opposite ends of the linear first oxidized interface layer **153***a*, as shown in FIG. **9**.
(29) Through the above-mentioned processes, the semiconductor device **100** according to the first embodiment of the present disclosure is formed. Accordingly to the fabricating method of the present embodiment, the semiconductor layer **142** entirely covering the substrate **110** is firstly formed, followed by defining the contact openings **146** through the semiconductor layer **142**, and forming the bit line contacts **160** in the contact openings **146**. The fabrication of the bit line contacts **160** is carried out by a multi-stepped deposition process, or by an epitaxial process combined with a deposition process, so that each of the bit line contacts **160** may therefor obtain a composite conductive layer, which includes the semiconductor layer **152***a*, the first conductive layer **154***a*, and the second conductive layer **156***a* stacked sequentially. It is noted that, the semiconductor layer **152***a*, the first conductive layer **154***a*, and the second conductive layer **156***a* may respectively include the same semiconductor material or different semiconductor materials, such as doped silicon, doped phosphorus, or silicon phosphide, and preferably all include silicon phosphide. Also, the first conductive layer **154***a* and the second conductive layer **156***a* disposed over the semiconductor layer **152***a* are preferably include a relative smaller thickness T**2**, for example being about 10-20 nanometers, so that, the top surface of each of the bit line contacts **160** may obtain a relative smaller grain size and a relative finer lattice structure, to present further smooth surface thereby.

(30) Furthermore, while the composite conductive layer is formed after forming the semiconductor material, through breaking vacuum or introducing oxygen, the lattice structure at the top of the semiconductor material is broken, and the first oxidized interface layer **153***a* is additionally formed between the semiconductor layer **152***a* and the first conductive layer **154***a*, and the second oxidized interface layer **155***a* is additionally formed between the first conductive layer **154***a* and the second conductive layer **156***a*. The first oxidized interface layer **153***a* includes a linear cross-section, and the second oxidized interface layer **155***a* still includes an U-shaped cross-section, and the first oxidized interface layer **153***a* and the second oxidized interface layer **155***a* are respectively disposed at the bottom surface and the top surface of the first conductive layer **154***a*, but not limited thereto. The thickness of the first oxidized interface layer **153***a* and the second oxidized interface layer **155***a* is quite thin, for example being about 0.01-1 angstroms. In this way, the existing of first oxidized interface layer **153***a* and the second oxidized interface layer **155***a* will not affect the electrical connection between each bit line **168** and each bit line contact **160**, and which may further improve the grain size and the lattice structure of the bit line contacts **160** (being more smooth), so as to improve the structural defects of the semiconductor material, such as excessive grain size and rough surface layer, caused by thicker stacked films. Then, the fabricating method of present disclosure is allowable to form the semiconductor device **100** with better performances under a simplified process flow.
(31) People in the art shall easily realize that the semiconductor device and the method of fabricating the same in the present disclosure are not limited to the aforementioned embodiment, and may include other examples. The following description will detail the different embodiments of the semiconductor device and method of fabricating the same in the present disclosure. To simplify the description, the following description will detail the dissimilarities among the different embodiments and the identical features will not be redundantly described. In order to compare the differences between the embodiments easily, the identical components in each of the following embodiments are marked with identical symbols.
(32) Please refer to FIG. **10** and FIG. **11**, which illustrate schematic diagrams of a semiconductor device **300** according to the second embodiment in the present disclosure. The forming process at the front end of the present embodiment is substantially the same or similar to those in the first embodiment for example being illustrated in FIGS. **1-5**, and those steps will not be redundantly described herein. The difference between the present embodiment and the aforementioned first embodiment is in that the contact openings of the present embodiment include a relative smaller diameter, and each of the bit line contacts **260** formed subsequently in the contact openings may therefore include the first oxidized interface layer **253***a* and the second oxidized interface layer **255***a* both in an U-shaped cross-section.
(33) Precisely speaking, as shown in FIG. **10**, a SEG process and a deposition process are sequentially performed on the substrate **110**, to form a first semiconductor material layer **252**, a second semiconductor material layer **254**, and a third semiconductor material layer **256** stacked from bottom to top, wherein the first semiconductor material layer **252**, the second semiconductor material layer **254**, and the third semiconductor material layer **256** for example include a semiconductor material like doped silicon, doped phosphorus or silicon phosphide, and preferably includes silicon phosphide, but not limited thereto. Also, as the semiconductor materials of the first semiconductor material layer **252**, the second semiconductor material layer **254**, and the third semiconductor material layer **256** are treated by breaking the vacuum or introducing oxygen, a first oxidized interface layer **253** is additionally formed between the first semiconductor material layer **252** and the second semiconductor material layer **254**, and a second oxidized interface layer **255** is additionally formed between the second semiconductor material layer **254** and the third semiconductor material layer **256**. The first oxidized interface layer **253** and the second oxidized interface layer **255** respectively include a quite thin thickness, for example being about 0.01-1 angstroms, and which will not affect the electrical connection between each bit line contact and

other elements formed subsequently. In one embodiment, the first oxidized interface layer **253** and the second oxidized interface layer **255** for example include silicon oxide, phosphorus oxide, or silicon phosphorus oxide, and preferably both include silicon phosphorus oxide, but not limited thereto.

(34) Then, an etching process and a deposition process are sequentially performed, to form a barrier material layer (not shown in the drawings), a metal material layer (not shown in the drawings), and a capping material layer (not shown in the drawings) stacked sequentially, after partially removing the third semiconductor material layer **256**, the second semiconductor material layer **254**, the first semiconductor material layer **252**, and the mask layer **140**. Following these, after performing the photolithography process, the semiconductor layer **142**, the barrier material layer, the metal material layer, and the capping material layer stacked sequentially are patterned to form a plurality of bit lines **268**. As shown in FIG. **11**, each of the bit lines **268** includes a semiconductor layer **242***a*, a barrier layer **262***a*, a metal layer **264***a*, and a capping layer **266***a* stacked from bottom to top. It is noted that, since the bit line contacts **260** each is formed under the barrier layer **262***a* of a corresponding bit line **268**, the semiconductor layer **242***a* of the corresponding bit line **268** is not illustrated in the cross-sectional view as shown in FIG. **11**, and the semiconductor layer **242***a* of the corresponding bit line **268** is formed at two sides of the bit line contacts **260** in the x-direction as being viewed from a top view (not shown in the drawings). It is noted that, the bit line contacts **260** are disposed under a portion of the bit lines **268**, to further extend into the active areas **114** of the substrate **110** to directly contact thereto. Furthermore, the bit line contacts **260** are also patterned through the photolithography process, to partially remove the first semiconductor material **252** disposed at two sides of each bit line contact **260**. Accordingly, each of the bit line contacts **260** may include a composite conductive layer, which includes the semiconductor layer **252***a*, the first oxidized interface layer **253***a*, the first conductive layer **254***a*, the second oxidized interface layer **255***a*, and the second conductive layer **256***a* stacked sequentially, with the first oxidized interface layer **253***a* and the second oxidized interface layer **255***a* both having an U-shaped cross-section, as shown in FIG. **11**.

(35) After that, as shown in FIG. **11**, a deposition process and an etching back process are sequentially performed, to form a plurality of bit line spacers **270** on two opposite sidewalls of each of the bit lines **268**, respectively. It is noted that, a portion of the bit line spacers **270** further extends into active areas **114** of the substrate **110** to cover on sidewalls of each of the bit line contacts **260**, and the portion of the bit line spacers **270** may directly contact two opposite ends of the U-shaped first oxidized interface layer **253***a*, as shown in FIG. **11**.

(36) Through the above-mentioned processes, the semiconductor device **300** according to the second embodiment of the present disclosure is formed. Accordingly to the fabricating method of the present embodiment, the formation of the bit line contacts **260** is also carried out by an epitaxial process combined with a deposition process, so that each of the bit line contacts **260** may therefor obtain a composite conductive layer, which includes the semiconductor layer **252***a*, the first conductive layer **254***a*, and the second conductive layer **256***a* stacked sequentially. It is noted that, the semiconductor layer **252***a*, the first conductive layer **254***a*, and the second conductive layer **256***a* may respectively include the same semiconductor material or different semiconductor materials, such as doped silicon, doped phosphorus, or silicon phosphide, and the first oxidized interface layer **253***a* is additionally formed between the semiconductor layer **252***a* and the first conductive layer **254***a*, and the second oxidized interface layer **255***a* is additionally formed between the first conductive layer **254***a* and the second conductive layer **256***a*, with the first oxidized interface layer **253***a* and the second oxidized interface layer **255***a* both including an U-shaped cross-section. In this way, the existing of first oxidized interface layer **253***a* and the second oxidized interface layer **255***a* will not affect the electrical connection between each bit line **268** and each bit line contact **260**, and which may further improve the grain size and the lattice structure of the bit line contacts **260** (being more smooth), so as to improve the structural reliability of the

semiconductor device **300**, and to promote the device performance thereby.
(37) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A semiconductor device, comprising: a substrate; an insulating layer, disposed on the substrate; a plurality of bit lines, disposed on the substrate; and a bit line contact, disposed between a corresponding one of the plurality of bit lines and the substrate, to electrically connect the corresponding one of the plurality of bit lines, wherein the bit line contact comprises a first conductive layer and a first oxidized interface layer, and a bottommost surface of the first oxidized interface layer is lower than a top surface of the insulating layer, wherein the bit line contact further comprises a second oxidized interface layer, and the second oxidized interface layer and the first oxidized interface layer are respectively disposed on a top surface and a bottom surface of the first conductive layer.
2. The semiconductor device according to claim 1, wherein the first oxidized interface layer comprises a linear cross-section.
3. The semiconductor device according to claim 1, wherein the first oxidized interface layer comprises an U-shaped cross-section.
4. The semiconductor device according to claim 1, wherein the second oxidized interface layer and the first oxidized interface layer respectively comprise an U-shaped cross-section and a linear cross-section.
5. The semiconductor device according to claim 1, wherein the second oxidized interface layer and the first oxidized interface layer both comprise an U-shaped cross-section.
6. The semiconductor device according to claim 1, wherein the bit line contact further comprises a second conductive layer, and the second oxidized interface layer is disposed between the first conductive layer and the second conductive layer, and the first conductive layer and the second conductive layer comprise a same material and a same thickness.
7. The semiconductor device according to claim 1, wherein the bit line contact further comprises a semiconductor layer, and the first oxidized interface layer is disposed between the first conductive layer and the semiconductor layer, wherein a thickness of the semiconductor layer is greater than a thickness of the first conductive layer.
8. The semiconductor device according to claim 7, wherein a grain size of the semiconductor layer is greater than a grain size of the first conductive layer.
9. The semiconductor device according to claim 7, wherein the first conductive layer and the semiconductor layer comprise a same material and different lattice structures.
10. The semiconductor device according to claim 1, wherein the bottommost surface of the first oxidized interface layer is lower than a bottommost surface of the insulating layer.
11. A semiconductor device, comprising: a substrate; an insulating layer, disposed on the substrate; a plurality of bit lines, disposed on the substrate; and a plurality of bit line contacts, each disposed between a corresponding one of the plurality of bit lines and the substrate, to electrically connect the corresponding one of the plurality of bit lines, wherein each of the plurality of bit line contacts comprises a first conductive layer and a first oxidized interface layer, and a bottommost surface of the first oxidized interface layer is lower than a top surface of the insulating layer, and a topmost surface of the first oxidized interface layer is higher than a top surface of the insulating layer.